AARKI

The Analytics Challenge

1. Introduction:

This analysis covers ~3000 leads generated through online advertising for a debt settlement company from April to September 2009 . Each row represents a lead generated through an ad campaign.

The goal of this analysis is to evaluate lead quality trends, identify drivers of lead quality, and explore opportunities to increase lead quality by 20%.

BUSINESS CONTEXT:

In the dataset Aarki acts as a Lead Generating Company which sells leads to an advertising company.The advertiser is a debt settlement company which buys good quality leads.

- Evaluate whether lead quality is improving or declining over time
- Identify which segments, ad creatives, and lead characteristics drive higher quality
- Determine whether lead quality can be improved from 8.0% to 9.6% to unlock a 20% CPL increase ($30 → $33)
- Evaluate if WidgetName affects lead quality.

A lead is a person who submitted their contact details after seeing an ad.

Understanding the Lead Funnel:

Customer journey:

ad clicked → user fills the form → advertiser calls lead→ lead status is assigned → customer joins the program .

Higher debt → **higher probability of closing**.

2. Data Cleaning:

1. Assigning 'Unknown' status to the blank CallStatus rows.

2. New column to separate Targeted ads ve generic ads.

   =IF(TRIM(L2)="Debt Settlement1 Master", "Generic Ad", "Targeted Ad")

AdvertiserCampaignName	Target/Generic
Debt Settlement1 Master	Generic Ad
Debt Settlement1 Master	Generic Ad
creditsolutions-branded-shortform	Targeted Ad

3. Add MonthYear column from LeadCreated using
   =TEXT(A2,"MMMM-yyyy")

LeadCreatedformatted	MonthYear
02-04-2009	April-2009
02-04-2009	April-2009
03-04-2009	April-2009

4. **LeadQuality column from CallStatus:**

   =IF(OR(F2="Closed", F2="EP Confirmed", F2="EP Sent", F2="EP Received"),1,0)
   If the status is either of the following it will be assigned 1 else 0:
   Closed/ EP Confirmed/ EP Sent/ EP Received.
   Otherwise it is 0.

   1 = good lead
   0 = bad/neutral lead

F	G
CallStatus	**LeadQuality**
Unable to contact - Bad Contact Information	0
Unknown	0
Closed	1
Closed	1
EP Confirmed	1
Closed	1

Leads Quality % = (Good Leads / Total Leads)*100

5. **LeadCategory: Map the CallStatus(Leads) into categories:**
   Good Lead → Closed, EP Confirmed, EP Received, EP Sent
   Bad Lead → Unable to contact - Bad Contact Information, Contacted - Invalid Profile, Contacted - Doesn't Qualify
   Unknown → Unknown

CallStatus	LeadQuality	LeadCategory
Unable to contact - Bad Contact Infor	0	Bad Lead
Unknown	0	Unknown
Closed	1	Good Lead
Closed	1	Good Lead
EP Confirmed	1	Good Lead
Closed	1	Good Lead
Unknown	0	Unknown

6. **DebtBucket:**

   This was the DebtLevel, where 7500–15000 is an overlapping range.

   Rather than mapping the ranges in a very long IF condition or manually doing it, we can make a temporary table and use XLOOKUP to get the data faster .

Range	DebtCat
7500-15000	Low
7500-10000	Low
10001-15000	Low
15001-20000	Low
20001-30000	Medium
30001-50000	Medium
50001-70000	High
70001-90000	High
90000-100000	Very High
More_than_100000	Very High

=XLOOKUP(Q2,'page 1'!P:P,'page 1'!Q:Q)

DebtLevel	DebtBucket
More_than_100000	Very High
7500-15000	Low
30001-50000	Medium
7500-15000	Low
7500-15000	Low
7500-15000	Low
30001-50000	Medium

7. **NULL Values in Address and Phone column:**

8. **VendorLeadId name issue:**
There are a few rows in the VendorLeadId that looks corrupted. There rows have the other data so for temporary , we will name them VendorLeadidtemp1, VendorLeadidtemp2 etc.

VendorLeadID
#NAME?
#NAME?
#NAME?

to

VendorLeadID
VendorLeadIDtemp1
VendorLeadIDtemp2
VendorLeadIDtemp3
VendorLeadIDtemp4
VendorLeadIDtemp5

9. **Change the widget size from 300250 to 300252 using** =SUBSTITUTE(J2,"300250","300252")
10. **Add a FormPage columns . If 1DC then 1 , 2DC is 2 pages.**

3. Exploratory Data Analysis (EDA) :

- **The Overall Lead Quality id 13% (more than 8%).**

LeadQuality% is the avg of the LeadQuality.

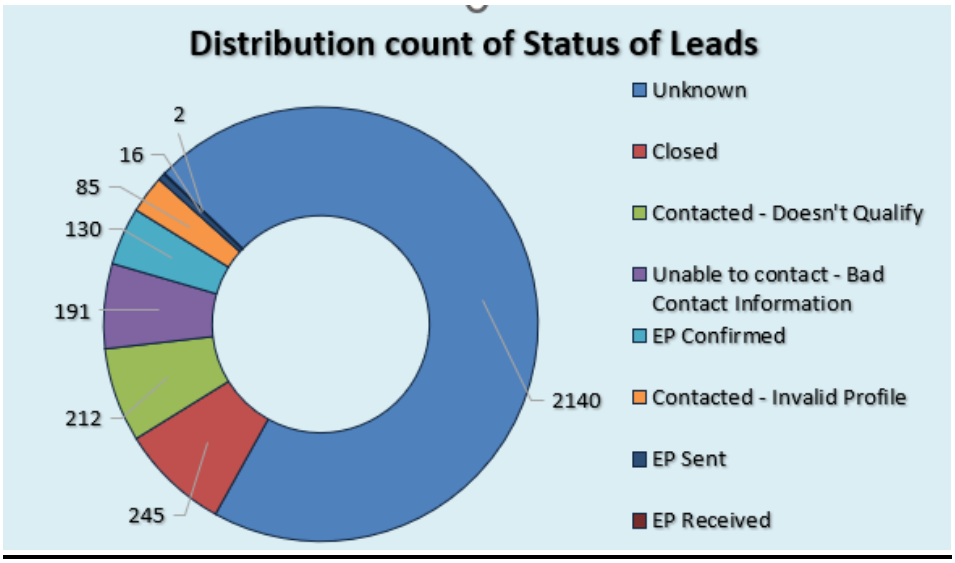
Count of VendorLeadID	LeadQuality%
3021	**13%**

This means approximately 13 out of every 100 leads generated eventually progress to meaningful engagement stages or conversion.

- **Distribution of CallStatus:**

CallStatus	Total
Unknown	2140
Closed	245
Contacted - Doesn't Qualify	212
Unable to contact - Bad Contact Information	191
EP Confirmed	130
Contacted - Invalid Profile	85
EP Sent	16
EP Received	2
Grand Total	**3021**

The highest CallStatus in 'Unknown', followed by 'Closed'



4. Analysis :

1. *Are we seeing any lead quality trends over time (improving, declining)? Are they statistically significant?*

Good leads are: Closed/ EP Confirmed/ EP Sent/ EP Received.

1 = good lead
0 = bad/neutral lead

NOTE: The recent leads haven't been closed yet.

MonthYear	Count of VendorLeadID	Sum of Good Lead	LeadQuality%	Average of LeadQuality	MoM Change
April-2009	518	79	15%	0.15	
May-2009	580	59	10%	0.10	-33.30%
June-2009	609	86	14%	0.14	38.82%
July-2009	532	49	9%	0.09	-34.78%
August-2009	438	62	14%	0.14	53.69%
September-2009	344	58	17%	0.17	19.11%
Grand Total	**3021**	**393**	**13%**	**0.13**	

Count of VendorLeadID is the total leads.

Sum of Good leads is the sum on the LeadQuality column.

The average of LeadQuality column is the LeadQuality%.

The Month over Month change in the LeadQuality is also shown.

Overall there was a decline if only considering the no. of good leads .

If we compare the first and the last months of the dataset there is a decline with a spike in good leads in the month of June.

If we check the LeadQuality % that is (good quality leads)/(total leads), we can see that there is an increasing trend here.

In September there is 17% of good lead quality of the total leads in that month.

To check for Statistical significance:

So even though the no. of Leads decreases over time the lead quality improves.

Lead quality percentage is trending upward(15% → 17%).

To verify this statistically ,

I have used =CORREL and got the output 0.270601514.

Weak positive correlation.

More data over a longer window would be needed to detect a reliable trend as the data I have currently used is only for 6 months.

2. *What can we learn about the drivers of "lead quality" from this dataset? What segments where the ad was shown, what kind of person filled out the ad, what kind of ad did they see have differing lead quality rates?*

- **Lead Quality by traffic source (Partner):**

Partner	Average of LeadQuality	Count of Leads
Advertise.com	33%	3
AdKnowledge	19%	171
Call_Center	16%	271
yahoo	14%	958
google	11%	1618
Grand Total	13%	3021

• Even though **Advertise.com shows the highest lead quality (33%)**, it has only **3 leads**, so this result is not reliable enough to draw conclusions.
• **AdKnowledge has strong lead quality at 19%**, which is great.
• **Call Center leads also perform well (16%)**.
• **Most partners have lead quality above the overall dataset average (13%)**.
High-performing sources like **AdKnowledge and Call Center** may attract users with stronger intent to resolve their debt.
• **Google generates a high volume of leads but lower quality**, likely because it is a broad search engine where users with lower intent may also search.
• **Advertise.com should be monitored further and evaluated once more leads are generated.**
• Increasing investment in **AdKnowledge and Call Center** could improve overall lead quality while maintaining moderate lead volume.
• **Google campaigns should be optimized**, as they generate the majority of the lead volume.

- **Lead Quality by Debt Categories:**

Debt settlement companies need **high debt customers**.

DebtBucket	Average of LeadQuality	Lead Count
High	17%	376
Medium	14%	952
Low	12%	1412
Very High	9%	281

- The highest lead quality % is from the 'HIGH' debt bucket(50001-90000) at 17% **%)**, indicating that users with larger debt amounts are more likely to seriously consider debt reduction programs .
- Next is 'MEDIUM' at 14%. Both of these are higher than the dataset average of 13%
- Very high debt leads(90000- more than 100000) have unexpectedly lower quality
- The highest volume is from the 'LOW' but have lower lead quality (12%)which might show lower urgency.
- Overall, **lead quality tends to increase from low to high debt levels**, showing that users with higher debt generally have stronger motivation to resolve their debt.
- Targeting **medium to high debt segments** could help improve overall lead quality, while also maintaining reasonable lead volume.

➢ Lead Quality by Target / Generic Ads:

Target/Generic	Average of LeadQuality	Lead Count
Generic Ad	13%	1815
Targeted Ad	13%	1206

- **Generic and targeted ads both show the same lead quality (13%),** which matches the overall dataset average.
- This is somewhat surprising, as targeted or branded ads are often expected to attract higher-intent users.
- Generic ads generate more leads (1815) compared to targeted ads (1206).
- **This suggests that ad type** does not significantly impact lead quality in this dataset, and other factors like traffic source or user intent may play a bigger role.

➢ Lead Quality by PublisherCampaignName:

PublisherCampaignName	Average of LeadQuality	Lead Count
DebtReductionCallCenter	16%	271
DebtReductionInc	13%	2750
Grand Total	**13%**	**3021**

- **Leads generated through the Call Center show higher lead quality (16%)** compared to online form leads (13%).
- However, **the majority of leads come from online submissions (2750**), while Call Center leads are much fewer (271).
- **This suggests that users who call directly may have stronger intent, but online** forms remain the main source of lead volume.

➢ Lead Quality by PhoneScore:

PhoneScore	Average of LeadQuality	Count of Leads
5	16%	619
NULL	13%	1628
3	12%	427
2	10%	162
4	7%	184
1	0%	1
Grand Total	**13%**	**3021**

•**Leads with a PhoneScore of 5 show the highest lead quality (16%)**, indicating that leads with strong verification tend to perform better.

• **A large portion of leads (1628) have NULL PhoneScore**, as this data was only collected recently, so these leads were not verified.

• **Lower PhoneScores (2 and 3) show lower lead quality**, suggesting that weaker matches between name and phone number may indicate lower quality leads.

- **Lead Quality by AddressScore:**

AddressScore	Average of LeadQuality	Count of Leads
2	18%	38
5	14%	870
NULL	13%	1850
1	11%	122
3	10%	93
4	4%	48

• **AddressScore 2 shows the highest lead quality (18%)**, but the sample size is very small (38 leads).
• **AddressScore 5 has strong lead quality (14%)** with a much larger number of leads (870).
• **Most leads (1850) have NULL AddressScore**, as this data was collected only for more recent leads.

- **Lead Quality by no. of form pages:**

(1DC→ 1 Page, 2DC→ 2 Page)

FormPages	Average of LeadQuality	Count of Leads
1	13%	2859
2	12%	162

• **Most leads come from 1-page forms (2859)** with a lead quality of **13%**.
• **2-page forms generate very few leads (162)** and show a slightly lower lead quality (**12%**).
• Overall, **the number of form pages does not appear to significantly impact lead quality** in this dataset.

➢ Lead Quality by Widget Name:

WidgetName	Average of LeadQuality	Count of Leads
w-302252-DebtReduction1-1DC-yellowarrow	24%	49
w-300252-DebtReduction1-1DC-CreditSolutions	23%	77
w-300252-DebtReduction1-1DC-white	17%	6
w-300252-DebtReduction1-1DC	16%	348
w-300252-DebtReduction1-1DC-Head2	16%	89
w-300252-DebtReduction1-1DC-BlueMeter	15%	92
w-300252-DebtReduction1-2DC-BlueMeter	14%	87
w-302252-DebtReduction1-1DC-yellowarrow-blue	13%	232
w-302252-DebtReduction1-1DC-CreditSolutions	12%	1054
w-302252-DebtReduction1-1DC	12%	272
w-302252-DebtReduction1-1DC-yellowarrow-dark	11%	135
w-300252-DebtReduction1-2DC-CreditSolutions	11%	75
w-302252-DebtReduction1-1DC-white	10%	430
w-300252-DebtReduction1-1DC-Head3	7%	75
Grand Total	**13%**	**3021**

Widget w-300250 and w-302252 are the same size so we have grouped them together. There are only 2 widgets with 2 page forms.

•**CreditSolutions widgets perform strongly**, with *w-300252-DebtReduction1-1DC-CreditSolutions* showing **23% lead quality**, one of the highest among widgets.

• The **yellowarrow widget (24%) also shows high lead quality**, though it has a relatively small number of leads.

Segments with differing lead quality rates:

• **Traffic Source (Partner):** AdKnowledge ~**19%** lead quality vs dataset average **13%**; Google generates high volume but lower quality.

• **Debt Level:** High debt (**50k–90k**) shows ~**17%** lead quality vs **~12%** for low-debt users.

• **Lead Channel: Call Center leads ~16%** vs **~13%** for online form leads.

• **Ad Creative (WidgetName): CreditSolutions ~23%** and **YellowArrow ~24%**, significantly above the **13% average**.

• **Phone Verification: PhoneScore = 5 → ~16%** lead quality, higher than lower scores.

3. Can we get to 9.6% quality? What would we do?

In the previous question we found the major drivers for increasing lead quality.

Strategy 1: Prioritize partner

Increase share of AdKnowledge(19%) and Call Center(16%) as they could improve overall lead quality while maintaining moderate lead volume.
Google campaigns should be optimized, as they generate the majority of the lead volume but has lower quality.

Strategy 2: Target High debt users:

Ads can be optimized to target consumers with higher debt ranges(5000-9000)(17%)vs lower debt(12%), as these segments demonstrate higher lead quality. People with more debt have stronger urgency to act.

Strategy 3: Improve lead verification quality for phone:

Leads with **PhoneScore = 5 show ~16% lead quality**, compared to lower scores which perform worse. Prioritizing leads with stronger phone verification may increase conversion probability. If the phone number doesn't match the person, the advertiser can't reach them, and the lead goes nowhere.

Only pass leads with PhoneScore ≥ 3 to the advertiser. Reject or reprice low-score leads.

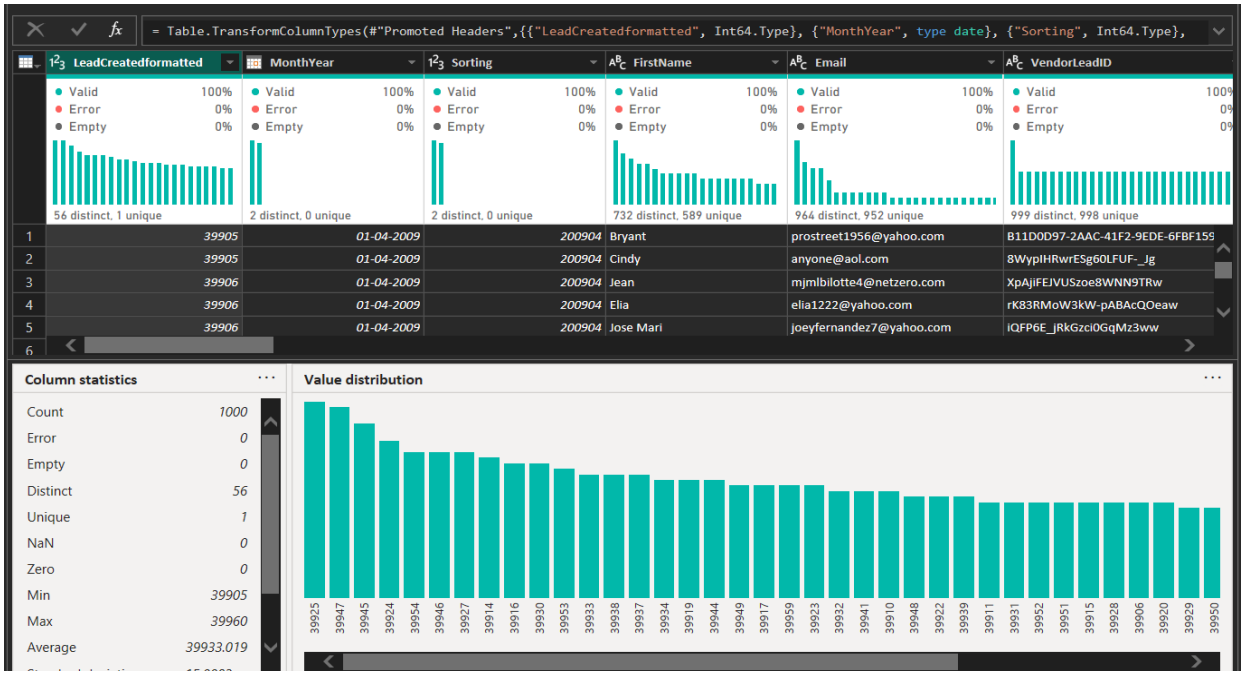
4. Evaluate if WidgetName affects lead quality.

- CreditSolutions widgets show ~23% lead quality, which is ~10 percentage points higher than the overall dataset average (13%).
- YellowArrow widgets show ~24% lead quality, also significantly above the dataset average, suggesting that certain ad designs perform better in attracting qualified leads.

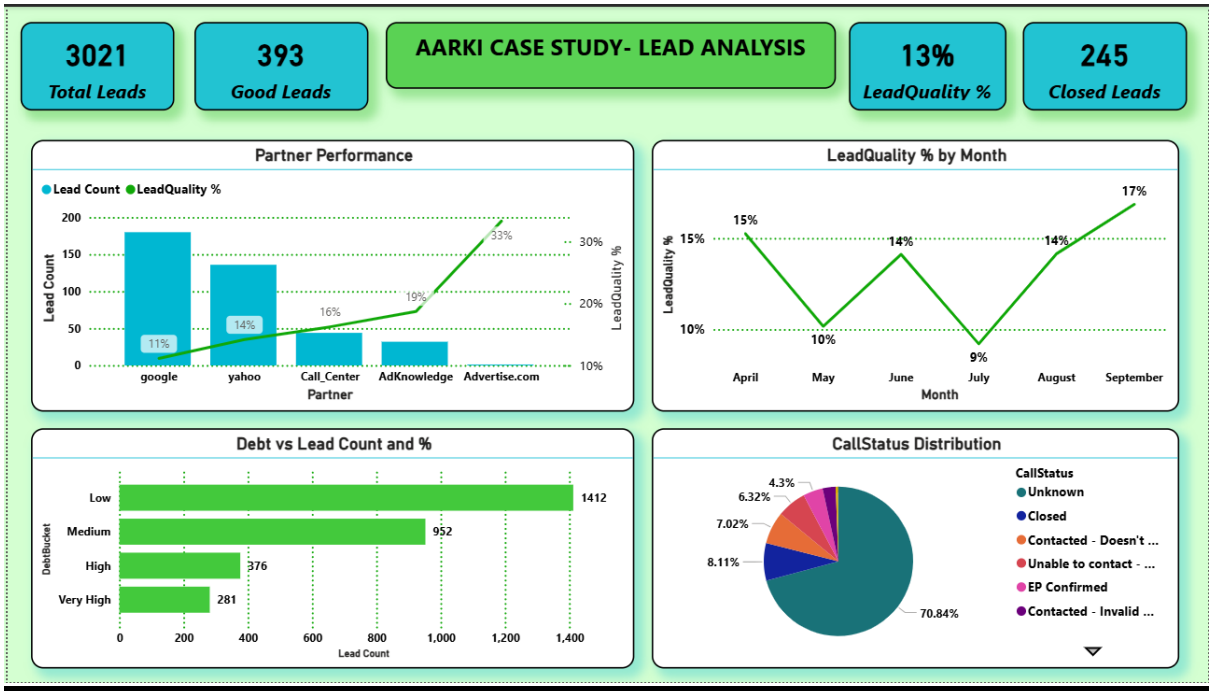
- Expanding the use of high-performing widgets like CreditSolutions and YellowArrow across campaigns could help improve overall lead quality.

#Data Visualisation in Power BI:

Import the excel file into Power Bi and transform the data.



Visualise the data:



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